# I. Agriculture in the 21st C.

## *Ag's Challenges and Solutionsg*

1. **Solutions**

   (a) remote sensing using multispectral images is a proxy for extensive manual crop monitoring operations in the field and provides potential savings in precious time and resources. Precision agriculture has grown and established itself as a central tool in agricultural management to forecast yield, monitor plant stress, and optimize fertilization, irrigation, and soil tilling activities for a more sustainable management of farming practices and improvements in yield for farmers.(Segarra et al., 2020)

## *Introduction to Remote Sensing in Ag*

1. **Satellites, Data, and Maps**

   (a) Satellites

      i. Sentinel-2

         A. the pivotal sensor: by providing freely accessible multispectral imagery with 10–20 m spatial resolution, frequent revisit times, and red-edge bands, Sentinel-2 has enabled a shift from regional assessments toward field-level and within-field yield monitoring, and its red-edge bands are also effective for monitoring nitrogen status. The Sentinel-2 constellation, with improved spatial, spectral, and temporal resolution was designed specifically to meet the needs of the agricultural community, both farmers and academic researchers with a focus on international agricultural development. (Segarra et al., 2020)

         B. One very notable use of Sentinel-2 features at field monitoring is the ESA Sen2-Agri system which generates modules of time series along the growing season. These different products consist of a monthly cloud-free composite of surface reflectance at 10–20 m resolution; monthly dynamic cropland masks, delivered from the agricultural mid-season onwards; a cultivated crop type map at 10 m resolution for main crop groups; and time series of vegetation status indicators, NDVI and LAI (Segarra et al., 2020)

      ii. Landsat 8

   (b) Satellite Data

      i. Concerning agriculture, in situations where crops interact with any given aspect of their environment (seasonal climatic variations, meteorological extreme events, pests, soil properties, etc.) or as crops grow and pass

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through different phenological stages, the interactions between plants and light reflectance translate into changes in plant signal patterns that can be interpreted using satellite data.(Segarra et al., 2020)

(c) Maps from Satellite Data

i.

(d) Applied Geospatial Analysis from Satellite Data

i.

2. **Drones, Data, and Maps**

(a) Drones

i.

(b) Drone Data

i.

(c) Maps from Drone Data

i.

(d) Applied Geospatial Analysis from Drone Data

## II. AI in the 21st C.

*Introduction to Professional AI Use*

1. **AI, APIs, and SDks**

## III. Records Management in the 21st C.

*Introduction to Records and Document Management*

1. **QAQC for Docs**

## IV.

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## References

Segarra, J., Buchaillot, M. L., Araus, J. L., & Kefauver, S. C. (2020). Remote Sensing for Precision Agriculture: Sentinel-2 Improved Features and Applications. *Agronomy*, *10*(5) (cit. on pp. 1, 2).

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